



Measuring the speed of light in a coaxial cable through differing cable lengths

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The coaxial cable has historically been a staple of rapid communication around the globe. As demand for speed and bandwidth increases, cable companies have been trying to keep up by optimizing for these things. A simple experiment was performed to measure the speed of light in a cable, finding a result that agrees with commonly accepted values. Due to limited time and materials, no further analysis could be made.

I. INTRODUCTION

A coaxial cable is a common type of electrical cable, consisting of layered copper and insulating materials. The first similar precursors were intended for telegraph signals, and the outer layers of copper served no purpose other than to shield the inner, signal-carrying conductor. In 1929, the first modern coaxial cables were patented, where the outside copper layers carried signal. These cables were intended to carry high frequencies, and much of the modern electrical world has been built upon their foundations.

With the advent of fiber optic and category cables, coaxial cables have been declining in popularity due to their limited bandwidth. Many advancements are being made to improve the efficiency of coaxial cables, including increased and higher quality shielding [1]. These changes aim to optimize the speed of the signal, a light wave, within the cable.

II. ANALYSIS

The speed of light in a vacuum is a well-known constant, $c_{\text{vac}} = 0.2998 \text{ m/ns}$. The speed of light changes, however, when light is propagating through a medium, due to the electric permittivity ϵ and magnetic permeability μ of the material through the relation

$$c = \frac{1}{\sqrt{\epsilon\mu}}. \quad (1)$$

The speed can easily be measured directly by taking

$$c = \frac{d}{t}, \quad (2)$$

where d is the distance traveled and t is the time taken to travel the distance.

An experiment was performed to utilize these equations and determine the speed of light in coaxial cables. Two cables of different lengths were connected to a signal

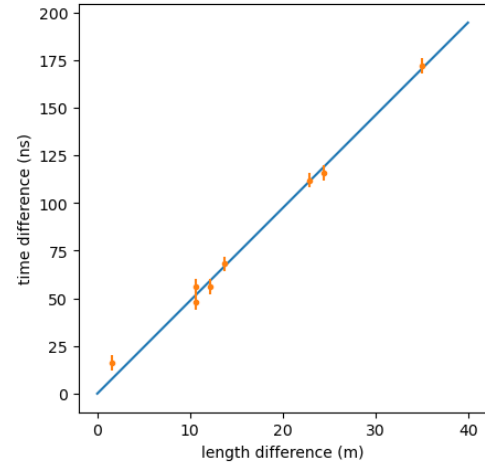


FIG. 1. The measured phase shift for cables of differing lengths. The slope represents the inverse of the speed of light in the cable, c . The c value was calculated to be $0.21 \pm 0.01 \text{ m/ns}$.

generator and an oscilloscope. The signal generator was set to produce a sine wave at a constant frequency of 2.2 MHz. Due to the available cables, a limited number of length differences were obtainable. For each cable difference, a phase difference was measured from the oscilloscope. This phase difference indicates the time it took for the light to travel through the excess cable, giving us everything we need to calculate the speed of light within the cable (fig. 1).

The speed of light in the coaxial cable was found to be $0.21 \pm 0.01 \text{ m/ns}$, found by fitting the line

$$t = \frac{d}{c} \quad (3)$$

to the data. Uncertainty calculations are detailed in Appendix A. Accepted values reside around 0.20 m/ns , within the realm of our error [2]. The fit had a reduced χ^2 value of 1.0002, indicating a strong fit. Higher precision could be obtained with more data points across more length differences and a finer-tuned oscilloscope.

Having measured the speed of sound in the coaxial cable, an electric permittivity constant could be determined using eq. (1). Unfortunately, insufficient data was collected to determine the value of μ or ϵ , and the author does not feel comfortable making an assumption for either of these numbers.

III. CONCLUSION

A simple experiment was performed to determine the speed of light through a coaxial cable by examining the phase difference from differing cable lengths. The acquired value agreed with common values of about two-thirds the speed of light in a vacuum. To produce a cable with a higher speed, electric permittivity and magnetic permeability must be reduced. Further studies should measure and examine these constants to produce a more efficient cable.

APPENDIX A: UNCERTAINTY

Uncertainty for the time delay was 4 ns, the resolution of the oscilloscope. Error in the speed of light was calculated using

$$\sigma_c = \frac{\sum \frac{d}{t^2} \sigma_t}{N}, \quad (4)$$

where σ denotes error, c denotes the speed of light, d is the difference in length, and t is the phase difference. The reduced χ^2 value was calculated by taking

$$\chi^2 = \frac{\sum \frac{(t_{obs} - t_{exp})^2}{\sigma_t^2}}{N}. \quad (5)$$

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- [1] S. H. Lampen, in *SMPTE 2016 Annual Technical Conference and Exhibition* (SMPTE, 2016) pp. 2–21.
 - [2] L. Grace, Speed of signal through coaxial cable.